

Using Google Colab for Beginners: A Step-by-Step Guide

Google Colab, short for **Colaboratory**, is a free cloud service hosted by Google that supports **Machine Learning** and **Artificial Intelligence** research. It provides an environment where you can use **Python** to run scripts, write and execute code, and share your work with others. **Colab** runs code on Google's cloud servers, which can be a great resource if your laptop is not capable of handling the computational requirements of large models like CNNs.

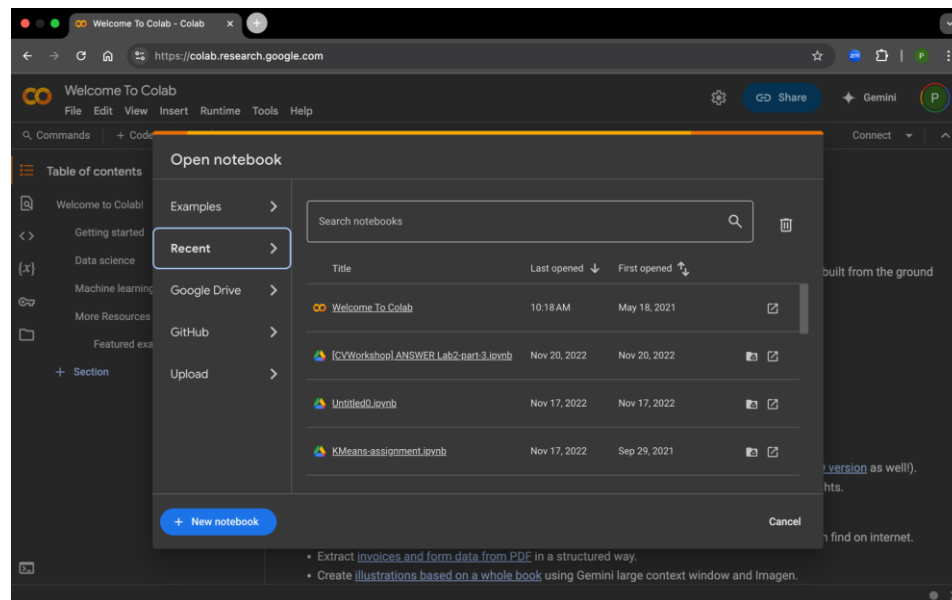
Using **Google Colab** means you are not running your code locally on your laptop, but instead on **Google's servers**. This allows you to leverage powerful resources such as GPUs and TPUs (which are ideal for deep learning tasks) without worrying about hardware limitations on your local machine. Here's a beginner's guide to using Google Colab:

Files:

1. data.csv
2. M13Lab.ipynb

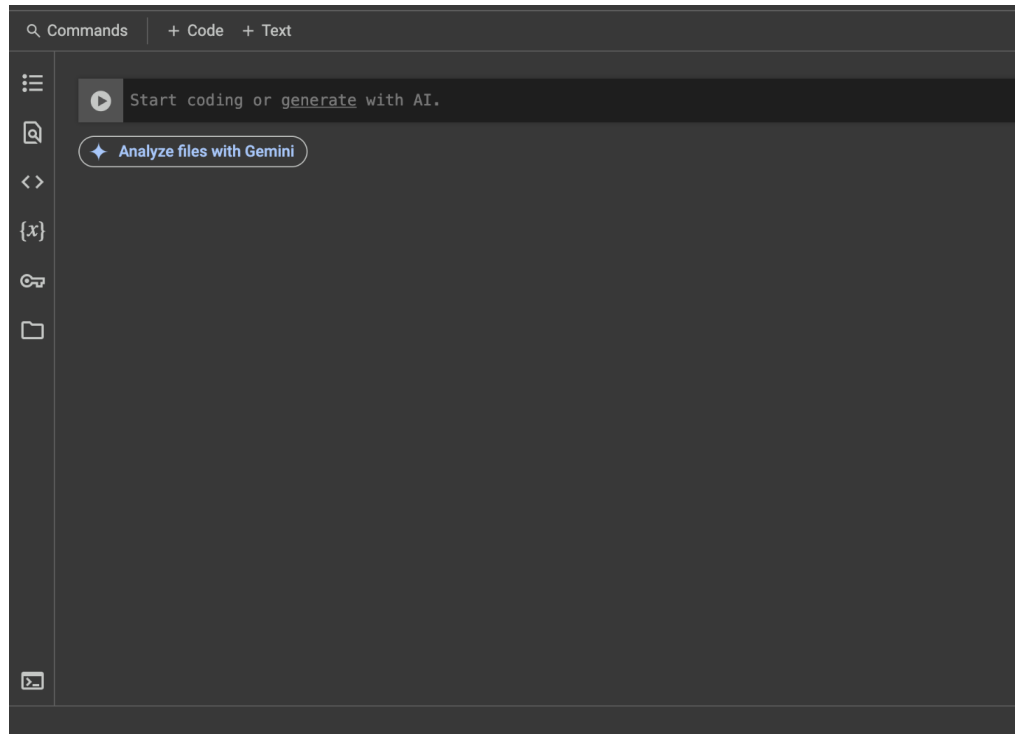
Step 1: Accessing Google Colab

- A. **Open Your Browser:** Start by opening your preferred web browser.
- B. **Navigate to Colab:** Go to [Google Colab](https://colab.research.google.com). If you are not signed into your Google account, you will be prompted to log in. You should see something like below:



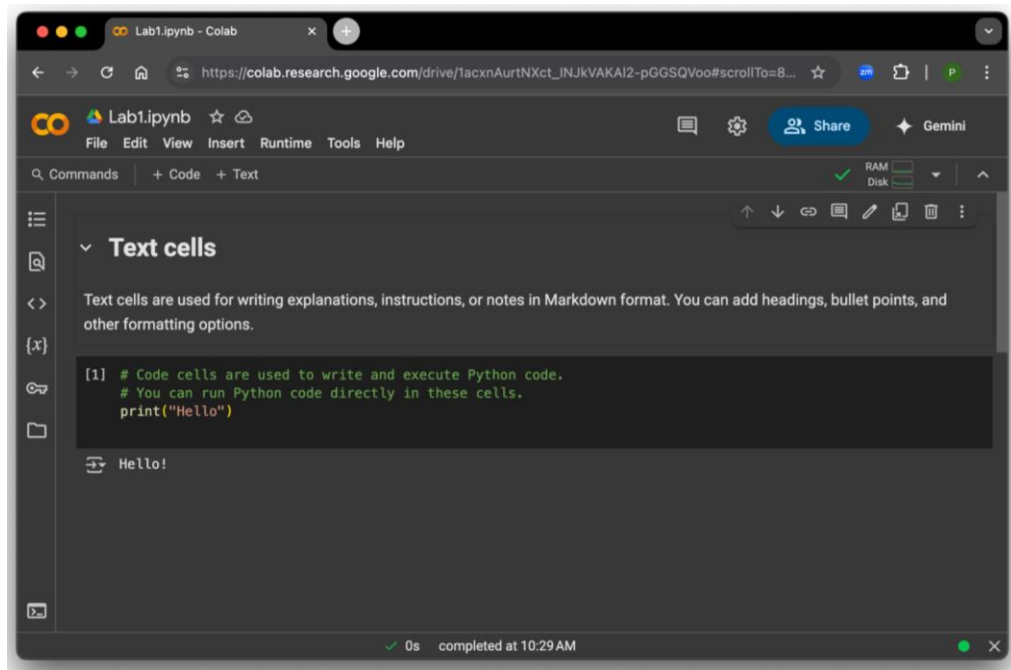
Step 2: Create a New Notebook

- A. **New Notebook:** Once you are on the Colab homepage, click on **New Notebook** which will open a new tab with a fresh notebook.
- B. **Rename the Notebook:** Click on the notebook title (usually "Untitled0.ipynb" by default) at the top to rename it to something more descriptive.



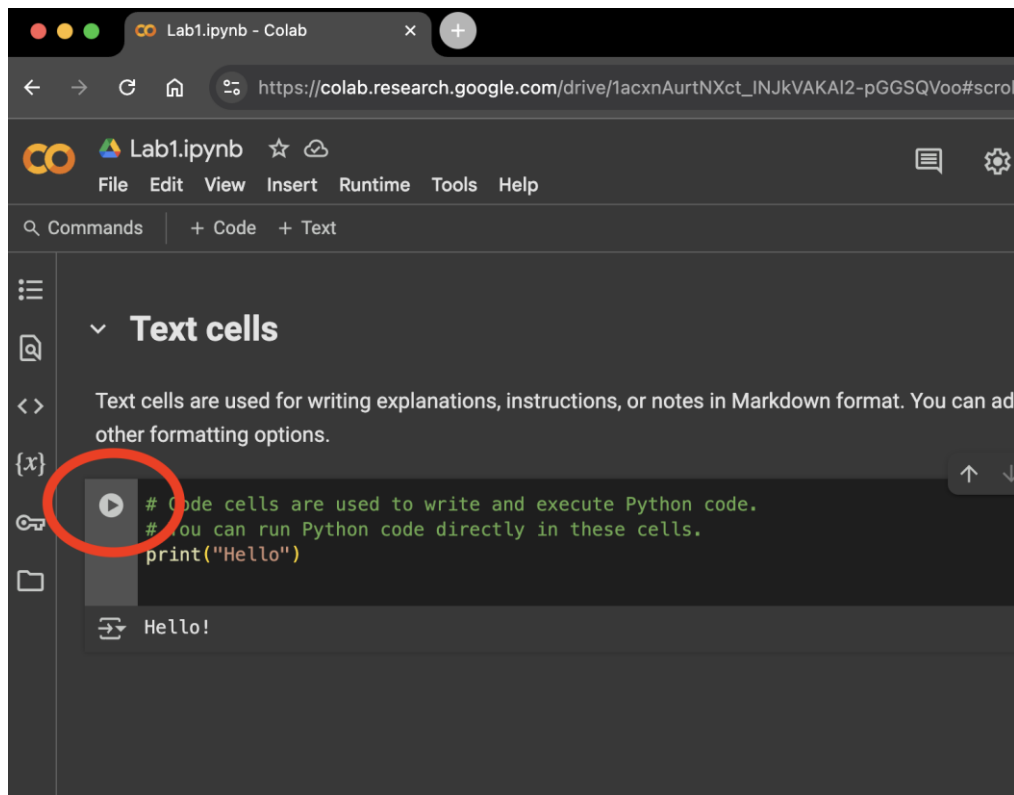
Step 3: Familiarizing With the Interface

- A. **Menu Bar:** At the top, you have the menu bar with options like File, Edit, View, Insert, Runtime, and Tools. These allow you to manage files, edit the notebook, change view settings, add cells, and configure the runtime environment.
- B. **Toolbar:** Below the menu bar, there's a toolbar with buttons for running cells, adding new cells, cutting, copying, pasting cells, and more.
- C. **Code and Text Cells:** The notebook consists of cells, which can be either code cells or text (markdown) cells. You can add explanations or instructions using text cells.



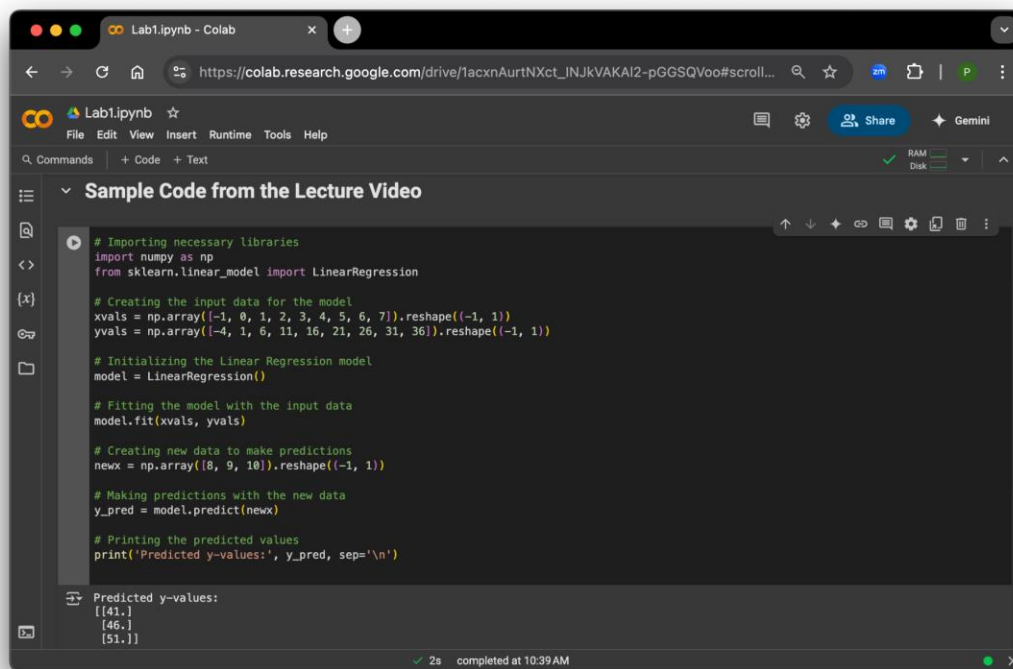
Step 4: Writing and Executing Code

- A. **Enter Code:** Click on a code cell or use **+ Code** to add a new code cell. Enter some Python code. For example, `print("Hello")`.
- B. **Run the Cell:** Press the play button to the left of the cell or use the shortcut **Ctrl+Enter** to execute the code. The output will appear directly below the cell.



Step 5: Importing Libraries and Installing Packages

- A. **Import Libraries:** Colab comes with many popular Python libraries pre-installed, such as **Numpy** for numerical operations and **Sklearn** for machine learning.
- B. You can import these libraries using standard Python syntax. For instance, you can import numpy to perform numerical operations or sklearn to utilize its built-in machine learning models.
- C. **Install Packages:** If you need additional packages that are not pre-installed in Colab, you can install them directly within your notebook.
- D. Use the pip install command followed by the package name to add new libraries to your Colab environment: `!pip install package-name`.
- E. The sample code in the lecture video to demonstrate how to perform linear regression analysis is shown below. The complete code is given in a separate Colab file. You may want to upload it to your Google drive and then open it.



```
# Importing necessary libraries
import numpy as np
from sklearn.linear_model import LinearRegression

# Creating the input data for the model
xvals = np.array([-1, 0, 1, 2, 3, 4, 5, 6, 7]).reshape((-1, 1))
yvals = np.array([-4, 1, 6, 11, 16, 21, 26, 31, 36]).reshape((-1, 1))

# Initializing the Linear Regression model
model = LinearRegression()

# Fitting the model with the input data
model.fit(xvals, yvals)

# Creating new data to make predictions
newx = np.array([8, 9, 10]).reshape((-1, 1))

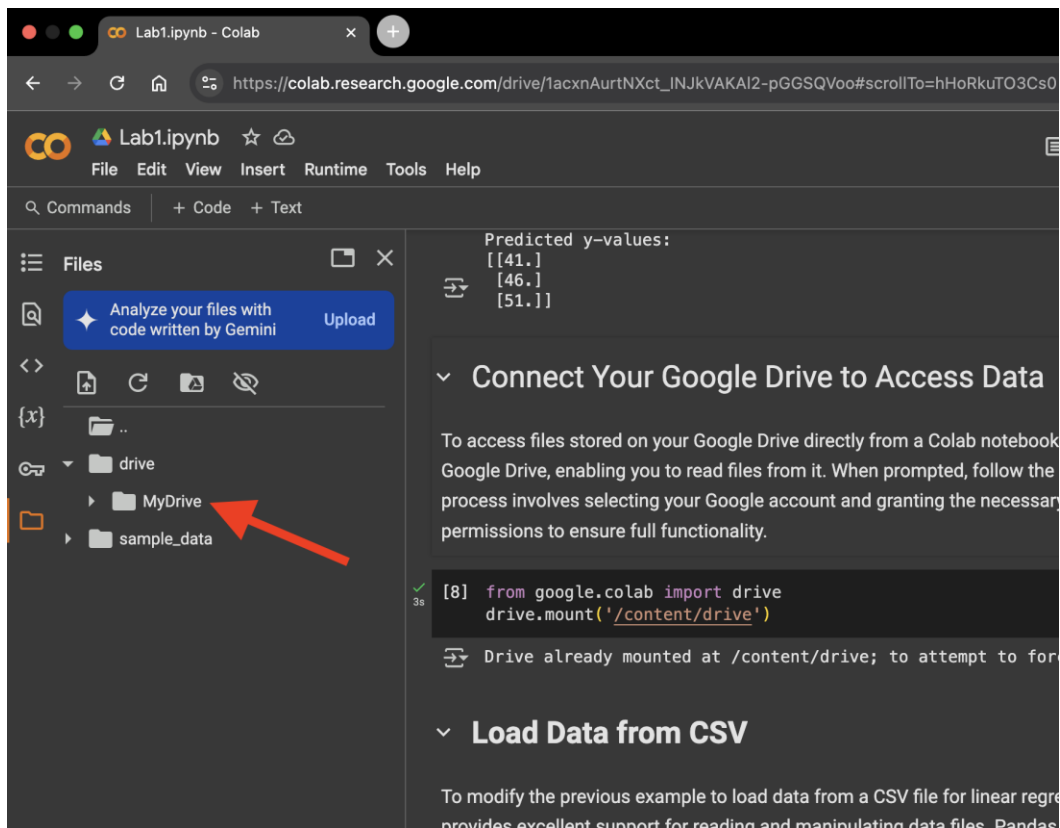
# Making predictions with the new data
y_pred = model.predict(newx)

# Printing the predicted values
print('Predicted y-values:', y_pred, sep='\n')
```

Predicted y-values:
[41.]
[46.]
[51.]

Step 6: Uploading and Using CSV file Data

- A. **Mount Google Drive:** To upload files directly to your Colab session, click on the folder icon located in the sidebar to open the file browser. Then, click on the Google drive icon to mount your drive to Colab. This will mount your Google Drive, enabling you to read files from it.
- B. When prompted, follow the on-screen instructions to authorize access. This process involves selecting your Google account and granting the necessary permissions. It's advisable to select all permissions to ensure full functionality.
- C. **Accessing Files from Google Drive:** Once you have mounted your Google Drive in your Colab notebook, it integrates seamlessly, allowing you to access and manage files. To find a specific file, like a CSV for data analysis, open the 'Files' tab on the left side of your Colab environment. Expand the 'drive' directory to enter into 'MyDrive', where your files are organized.
- D. **Copy the path:** Now, you should upload the given CSV file to your Google Drive and copy the file's path. This will allow you to directly access the file within your code cells by specifying its path.
- E. In the 'Files' sidebar, use the dropdown arrows next to the folder names to navigate through the directories. When you locate your desired file, right-click on it and select 'Copy path'. This provides the full path necessary to load the file directly into your notebook. Utilizing this path in your Python code allows you to read the file efficiently, integrating cloud storage into your data workflow smoothly.



Step 7: Loading Data from CSV for Linear Regression

- A. **Load Data:** In the previous Step 5, we hardcoded the x and y values as NumPy arrays directly in the code. Instead of hardcoding, we'll load x and y values from a CSV file using Pandas. Update the copied path to the `path` variable in the code below. This approach simulates real-world data handling and prepares us for working with larger datasets.
- B. **Train the Model:** We'll use the data loaded from the CSV to train our Linear Regression model, enabling us to understand the dynamics of dataset-driven training.
- C. **Predict New Values:** Finally, we'll predict y values for new x data, demonstrating the model's ability to generalize from the data provided in the CSV file.

```
# Importing necessary libraries
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression

path = '/content/drive/MyDrive/data.csv'

# Load data from a CSV file
# Ensure your CSV file is in the format of two columns: x values and y values without headers
data = pd.read_csv(path, header=None, names=['x', 'y'])

# Reshape data to fit the model: '-1' lets numpy determine the number of rows based on original data and number of columns
xvals = data['x'].values.reshape((-1, 1)) # Reshape x into a 2D array for sklearn
yvals = data['y'].values.reshape((-1, 1)) # Reshape y into a 2D array for sklearn

# Initialize the Linear Regression model
model = LinearRegression()

# Train the model with the dataset
model.fit(xvals, yvals)

# New data for prediction
newx = np.array([8, 9, 10]).reshape((-1, 1)) # New x values for which we want predictions

# Make predictions based on the new x values
y_pred = model.predict(newx)

# Print out the predicted y-values
print('Predicted y-values:', y_pred, sep='\n')

# Output will show predicted y-values for the input array [8, 9, 10]
```

Predicted y-values:
[[41.]
[46.]
[51.]]

Step 8: Saving and Sharing the Notebook

- A. **Save:** Your notebook automatically saves changes to Google Drive as you work. To manually save your progress, navigate to **File > Save**.
- B. **Share:** To share your notebook, click on **Share** in the upper right corner. Here, you can add the email addresses of collaborators or generate a link to share.
- C. **Download:** To download your notebook to your local machine, go to **File > Download > Download .ipynb**.

Step 9: Exploring Further

- A. Explore sample notebooks provided by Colab by visiting **File > Open notebook > Examples**.
- B. Learn more about markdown and advanced features in Colab notebooks to enhance your documentation.

Google Colab provides a powerful platform for anyone looking to delve into data science and machine learning without requiring local setup. It's an excellent tool for collaboration and sharing your work with peers or the public.